

Software Cost Estimation

Function Point Analysis & Use Case Points

CA Firm Attendance and Article Management System

Assignment No. 8

Group: E-17

Members: Nivesh Jain
Harsh Nayak
Rachit Ingole
Hasan Rupawalla

1. Problem Statement

Perform software cost estimation exercises using **Function Point Analysis (FPA)** and **Use Case Points (UCP)** for the CA Firm Attendance and Article Management System. Estimate the software size, effort, and approximate development cost based on these two industry-standard techniques.

2. Introduction

Software cost estimation is a fundamental activity in project management that predicts the resources, time, and budget required to develop a software system. Accurate estimation helps prevent budget overruns, enables proper resource allocation, and sets realistic delivery expectations.

Two widely used estimation techniques are:

- i) **Function Point Analysis (FPA)** — measures software size based on the functional requirements delivered to the user, independent of the programming language or technology used.
- ii) **Use Case Points (UCP)** — estimates effort based on the complexity of use cases and the environmental and technical factors of the project.

This assignment applies both techniques to the **CA Firm Attendance and Article Management System**, a web-based platform built using React.js, Node.js/Express, and MySQL that automates attendance tracking (biometric face recognition + GPS verification), task assignment, progress monitoring, leave management, and reporting for a Chartered Accountant firm (SPCM).

3. Function Point Analysis (FPA)

3.1 Overview

Function Point Analysis, introduced by Allan Albrecht at IBM in 1979, quantifies software functionality from the user's perspective. It classifies system interactions into five categories:

- **External Inputs (EI)** — data entering the system from outside.
- **External Outputs (EO)** — data or reports leaving the system.
- **External Inquiries (EQ)** — input/output combinations for data retrieval.
- **Internal Logical Files (ILF)** — data maintained within the system.
- **External Interface Files (EIF)** — data referenced from external systems.

3.2 Step 1: Identify and Classify Function Points

External Inputs (EI)

#	Function	Complexity
1	Admin Login	Low
2	Article Trainee Login	Low
3	Add Article Trainee (Admin)	Average
4	Edit Article Trainee Details (Admin)	Average
5	Mark Attendance (Face + GPS)	High
6	Create Task (Admin)	Average
7	Assign Task to Article(s)	Average
8	Update Task Progress (Article)	Average
9	Submit Leave Request (Article)	Low
10	Approve/Reject Leave (Admin)	Low
11	Configure System Settings (Admin)	Low

External Outputs (EO)

#	Function	Complexity
1	Attendance Report (Daily/Monthly)	High
2	Performance Report	High
3	Task Completion Statistics	Average
4	Leave Summary Report	Low
5	Analytics Dashboard (Chart.js)	High
6	Email Notifications (Nodemailer)	Average

External Inquiries (EQ)

#	Function	Complexity
1	View Attendance History	Low
2	View Assigned Tasks (Article)	Low
3	View All Articles List (Admin)	Low
4	View Leave Request Status	Low
5	View Task Progress (Admin)	Average
6	View Notification History	Low

Internal Logical Files (ILF)

#	Data Store (MySQL Table)	Complexity
1	admins	Low
2	articles (with face_descriptor)	High
3	attendance	High
4	tasks	Average
5	task_assignments	Average
6	progress_log	Average
7	leave_requests	Average
8	notifications	Low
9	settings	Low

External Interface Files (EIF)

#	External System	Complexity
1	face-api.js (Face Recognition)	High
2	Browser Geolocation API (GPS)	Average
3	Nodemailer (SMTP Email)	Low
4	Chart.js (Data Visualization)	Average

3.3 Step 2: Calculate Unadjusted Function Points (UFP)

The standard weighting factors are:

Type	Low	Average	High
External Inputs (EI)	3	4	6
External Outputs (EO)	4	5	7
External Inquiries (EQ)	3	4	6
Internal Logical Files (ILF)	7	10	15
External Interface Files (EIF)	5	7	10

Calculation:

Type	Low	Average	High	Total
EI	$4 \times 3 = 12$	$5 \times 4 = 20$	$1 \times 6 = 6$	38
EO	$1 \times 4 = 4$	$2 \times 5 = 10$	$3 \times 7 = 21$	35
EQ	$5 \times 3 = 15$	$1 \times 4 = 4$	—	19
ILF	$3 \times 7 = 21$	$4 \times 10 = 40$	$2 \times 15 = 30$	91
EIF	$1 \times 5 = 5$	$2 \times 7 = 14$	$1 \times 10 = 10$	29
Unadjusted Function Points (UFP)				212

3.4 Step 3: Value Adjustment Factor (VAF)

The VAF is calculated using 14 General System Characteristics (GSCs), each rated 0–5.

#	Characteristic	Rating (0–5)
1	Data Communications	4
2	Distributed Data Processing	2
3	Performance	4
4	Heavily Used Configuration	2
5	Transaction Rate	3
6	Online Data Entry	5
7	End-User Efficiency	4
8	Online Update	4
9	Complex Processing	5
10	Reusability	3
11	Installation Ease	3
12	Operational Ease	4
13	Multiple Sites	2
14	Facilitate Change	3
Total Degree of Influence (TDI)		48

$$\text{VAF} = 0.65 + (0.01 \times \text{TDI}) = 0.65 + (0.01 \times 48) = \mathbf{1.13}$$

3.5 Step 4: Adjusted Function Points (AFP)

$$\text{AFP} = \text{UFP} \times \text{VAF} = 212 \times 1.13 = \mathbf{239.56 \approx 240 \text{ FP}}$$

3.6 Step 5: Effort and Cost Estimation

Using industry averages for a web application project:

- Productivity rate: **10 FP per person-month** (moderate complexity MERN-equivalent stack)
- Average developer cost: **INR 40,000 per month** (academic/intern-level team)

$$\text{Effort} = \frac{\text{AFP}}{\text{Productivity}} = \frac{240}{10} = \mathbf{24.0 \text{ person-months}}$$

With a team of 4 developers:

$$\text{Duration} = \frac{24.0}{4} = \mathbf{6.0 \text{ months}}$$

$$\text{Total Cost} = 24.0 \times 40,000 = \mathbf{\text{INR } 9,60,000}$$

4. Use Case Points (UCP) Estimation

4.1 Overview

The Use Case Points method, proposed by Gustav Karner in 1993, estimates effort based on the complexity of actors and use cases, adjusted by technical and environmental factors.

4.2 Step 1: Unadjusted Actor Weight (UAW)

Actors are classified by their interaction type:

Actor	Type	Weight	Count
Admin (Chartered Accountant)	Complex (GUI-based)	3	1
Article Trainee	Complex (GUI-based)	3	1
face-api.js (Face Recognition)	Average (library/protocol)	2	1
Browser Geolocation API (GPS)	Simple (API)	1	1
Nodemailer (Email SMTP)	Simple (API)	1	1
Chart.js (Visualization)	Simple (API)	1	1
		UAW	11

4.3 Step 2: Unadjusted Use Case Weight (UUCW)

Use cases are classified by transaction count:

Use Case	Complexity	Weight	Trans.
Login / Logout	Simple (≤ 3)	5	3
Add Article Trainee	Average (4–7)	10	5
Edit/Remove Article	Simple	5	3
Mark Attendance (Face + GPS)	Complex (> 7)	15	12
View Attendance History	Simple	5	2
Create & Assign Task	Average	10	6
Update Task Progress	Average	10	5
Apply for Leave	Simple	5	3
Approve/Reject Leave	Simple	5	3
Generate Attendance Report	Complex	15	9
Generate Performance Report	Complex	15	8
View Analytics Dashboard	Average	10	7
Configure System Settings	Simple	5	2
		UUCW	115

4.4 Step 3: Unadjusted Use Case Points (UUCP)

$$\text{UUCP} = \text{UAW} + \text{UUCW} = 11 + 115 = \mathbf{126}$$

4.5 Step 4: Technical Complexity Factor (TCF)

T#	Factor	Wt.	Rating (0–5)
T1	Distributed System	2	3
T2	Response Time / Performance	1	4
T3	End-User Efficiency	1	4
T4	Complex Internal Processing	1	5
T5	Reusability	1	3
T6	Easy to Install	0.5	3
T7	Easy to Use	0.5	4
T8	Portability	2	3
T9	Easy to Change	1	3
T10	Concurrency	1	3
T11	Security Features	1	5
T12	Third-Party Access	1	4
T13	Training Required	1	2
TFactor = $\sum(\text{Wt.} \times \text{Rating})$			48.5

$$\text{TCF} = 0.6 + (0.01 \times 48.5) = \mathbf{1.085}$$

4.6 Step 5: Environmental Complexity Factor (ECF)

E#	Factor	Wt.	Rating (0–5)
E1	Familiarity with Project	1.5	3
E2	Application Experience	0.5	2
E3	OOP Experience	1	4
E4	Lead Analyst Capability	0.5	3
E5	Motivation	1	4
E6	Stable Requirements	2	4
E7	Part-Time Workers	–1	3
E8	Difficult Programming Language	–1	1
EFactor = $\sum(\text{Wt.} \times \text{Rating})$			19.0

$$\text{ECF} = 1.4 + (-0.03 \times 19) = 1.4 - 0.57 = \mathbf{0.83}$$

4.7 Step 6: Adjusted Use Case Points (UCP)

$$\text{UCP} = \text{UUCP} \times \text{TCF} \times \text{ECF} = 126 \times 1.085 \times 0.83 = \mathbf{113.4 \approx 113 \text{ UCP}}$$

4.8 Step 7: Effort and Cost Estimation

Using the standard productivity factor of **20 person-hours per UCP**:

$$\text{Effort (hours)} = 113.4 \times 20 = \mathbf{2,268 \text{ person-hours}}$$

Converting to person-months (assuming 160 working hours/month):

$$\text{Effort (months)} = \frac{2268}{160} = \mathbf{14.175} \approx \mathbf{14.2} \text{ person-months}$$

With a team of 4 developers:

$$\text{Duration} = \frac{14.2}{4} \approx \mathbf{3.6} \text{ months}$$

$$\text{Total Cost} = 14.2 \times 40,000 = \text{INR } \mathbf{5,68,000}$$

5. Comparison of Results

Parameter	FPA	UCP
Software Size	240 FP	113 UCP
Effort	24.0 person-months	14.2 person-months
Duration (4 developers)	6.0 months	3.6 months
Estimated Cost	INR 9,60,000	INR 5,68,000

The FPA method produces a higher estimate because it captures the complexity of all 9 MySQL database tables and the heavy internal processing involved in biometric face recognition (face descriptor comparison using Euclidean distance) and GPS-based geofencing (Haversine distance calculation). These internal computations significantly inflate the ILF and EI counts.

The UCP method yields a more conservative figure as it focuses on user-facing use cases and adjusts for the team's environment. The high ECF adjustment (due to stable requirements from a single-client CA firm) reduces the final estimate.

A realistic estimate for this project lies between the two bounds, approximately **19 person-months** of total effort, translating to roughly **4.7 months** of development with a 4-member team.

6. Conclusion

This assignment demonstrated two complementary approaches to software cost estimation for the CA Firm Attendance and Article Management System. Function Point Analysis provided a comprehensive size measurement accounting for all data interactions, biometric processing, and system boundaries, yielding 240 adjusted function points. The Use Case Points method offered an effort-centric estimate grounded in use case complexity and team environment factors, producing 109 adjusted use case points.

Together, these techniques provide upper and lower bounds for project planning, enabling the development team to set realistic milestones, allocate resources proportionally, and manage stakeholder expectations with data-driven estimates.